

Impact of Rayleigh backscattering on next-generation bidirectional 200 Gb/s FDMA coherent PON

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Abstract We experimentally evaluate the impact of Rayleigh backscattering in the upstream in a single-wavelength 200 Gb/s DP-QPSK FDMA coherent PON between two ONUs and an OLT. Simulations show a trade-off between downstream and upstream transmit powers and identify an operational region enabling simultaneous bidirectional transmission. ©2026 The Author(s)

Introduction

Point to multipoint (P2MP) passive optical networks (PON) enable rates up-to 50 Gb/s based on intensity modulated (IM) transmitters and direct detection (DD) receivers. The standardization process for 200 Gb/s very high speed PON (VHSP) is ongoing in the ITU-T study group 15 question 2 [1]-[3]. Since chromatic dispersion (CD) impact grows quadratically with symbol-rate (R_s), IM/DD VHSP systems would be restricted to the O-Band, creating coexistence challenges [3] and requiring complex digital signal processor (DSP) architectures [4]. An optical distribution network (ODN) with coherent transmission enables C-Band operation with DSP based CD pre- or post-compensation, and potential reuse of form factors from coherent transceivers. VHSP transmission has been demonstrated for IM/DD in [5],[6], and for coherent transmission in [7]-[9]. In bidirectional transmission, reflections from Rayleigh backscattering (RBS) impairs the signal within the same wavelength band. A single wavelength band for upstream (US) and downstream (DS) enables a single laser in the optical network unit (ONU) and optical line terminal (OLT), reducing cost [10]. Coherent transmission can increase tolerance to RBS, further increased if spectral separation is applied between US and DS [11]-[14], at the potential cost of higher bandwidth requirement in the transmitter and receiver.

In this work, we study the impact of RBS on a single wavelength frequency division multiple access (FDMA) coherent PON. We show experimentally the impact of RBS in a US 200 Gb/s FDMA transmission with two ONUs and an OLT. The DS transmission is emulated by ASE noise following a 60 GHz gain profile. In simulations we show a trade-off between DS (P_{Tx}^{DS}) and US (P_{Tx}^{US}) transmitting powers per sub-band. A proper selection of P_{Tx}^{DS} and P_{Tx}^{US} , enables simultaneous reception in the OLT and ONUs receiver, in an ODN with power splitter (PS) insertion loss (IL) of 23 dB (number of users $N_u=128$) and 35 dB of max. optical path loss (OPL) [15].

Experimental setup

Fig. 1 a) shows the setup used for experimental investigation of RBS impact on a US link. After generation of a pseudo-random bit sequence (PRBS), and mapping to dual polarization-quadrature phase shift keying (DP-QPSK) at 30 GBd in transmitter DSP (Tx-DSP) (Fig 2), two transmitters generate the FDMA single carrier signals under test with frequency separation (f_{sep}) of 2 GHz and pulse shaping root-raised cosine (RRC) roll-off factor of 5%. At Tx-DSP, header sequences are inserted for later frequency offset correction, frame synchronization, and channel estimation (CAZAC-64) in the data-aided frequency domain equalizer (DA-FDE [16]). Base-band signals are created by 8-bit, 120 GSa/s digital-to-analog converter (DAC, 35 GHz 3-dB bandwidth), and an external cavity laser (ECL) connected to dual polarization IQ modulators (DP-IQM, 40 GHz and 25 GHz 3-dB bandwidth) and erbium doped fiber amplifiers (EDFA) set both output powers to 4.5 dBm. ONU-1 and ONU-2, are connected through 11.4 km and 8.4 km standard single mode fiber (SSMF, drop fibers), followed by a 2:2 PS and 20 dB attenuation, emulating a total PS IL of 23 dB for $N_u=128$, higher than the ideal 21.07 dB due to considered additional IL. After 25 km SSMF (feeder fiber), the signal is received in the coherent receiver (CoRx) via port 3 of the circulator, prior to digitization at 256 GSa/s (10-bit ADC, 73 GHz 3-dB bandwidth). Digital subcarrier multiplexing (DSCM) receiver DSP (Rx-DSP, Fig. 2) extracts BER. DS link is emulated with an amplified spontaneous emission (ASE) noise source followed by a wavelength selective switch (WSS), that adds a 60 GHz gain profile, and an EDFA to amplify the noise power. Two optical spectrum analyzers (OSA) measure the normalized US and emulated DS spectrum at port 3 of the circulator or at the 2:2 PS (Fig. 1 b)). Variable optical attenuators (VOAs), VOA₁ and VOA₂, at port 1 of the circulator and at the CoRx input, respectively, enable

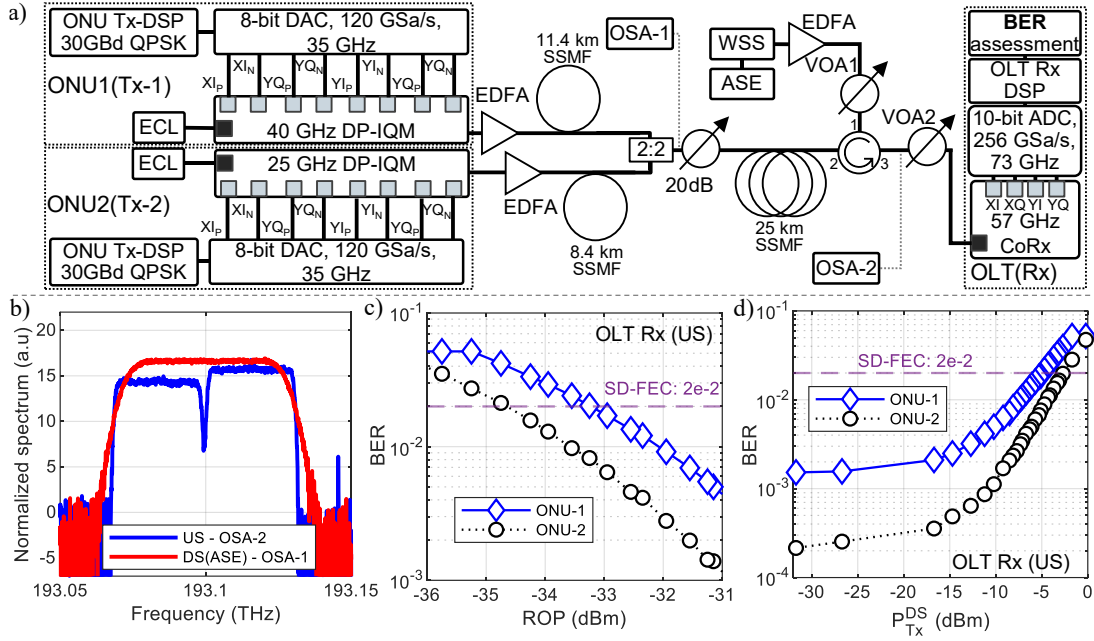


Fig. 1: (a) Experimental setup for Rayleigh backscattering impact assessment in the upstream. (b) normalized spectra measured by OSA-1 and OSA-2 for the upstream signal and downstream ASE in the emulated bidirectional link. (c) Sub-band received optical power (ROP) vs. BER in the OLT Rx-DSP (no DS). (d) P_{Tx}^{DS} vs BER in the OLT Rx-DSP ($VOA_2=0$ dB)

sweeps to assess the OLT Rx BER, versus received optical powers per subcarrier (ROP) for $VOA_1=100$ dB (No DS, Fig. 1 c)), and versus P_{Tx}^{DS} with $VOA_2=0$ dB (Fig 1 d)). Fig. 1 c) shows that the SD-FEC threshold of $2e-2$ [17] is met for ROPs above -33.25 dBm (ONU-1) and -34.65 dBm (ONU-2). Fig. 1 d) shows that P_{Tx}^{DS} degrades BER OLT Rx performance due to RBS, with maximum P_{Tx}^{DS} tolerance of -4.9 dBm (ONU-1) and -2.65 dBm (ONU-2).

Simulation setup

Fig. 2 shows the bidirectional US/DS simulation setup between two ONUs and the OLT, implemented in VPIphotonics™ Design Suite 11.6 with the VPIToolkit™ DSP Library 5.5. Simulation parameters follow the experimental evaluation. The US and DS lie within a 65.5 GHz single-wavelength band around $f_0 = 193.1$ THz. In the US, the ONU Tx-DSP generates a 30 GBd DP-QPSK single-carrier, with PRBS, QPSK mapping, 5% roll-off RRC pulse shaping and header insertion, producing two 65536 symbols frames, each with 992 symbols headers (1.54% header overhead)[16]. The sub-band frequencies of ONU-1 and ONU-2 are $f_1 = 193.084$ THz and $f_2 = 193.116$ THz with 2 GHz separation. The OLT and ONUs lasers have 100 kHz linewidth (LW), 1 GHz carrier frequency offset (CFO) and powers $p_f 14.5$ dBm and 16 dBm respectively. The ONU-1 and ONU-2 Tx-DSP drive 8-bit DACs with sampling rate $f_s=120$ GSa/s, and DP-IQMs with fourth-order Butterworth response of 35 GHz and 25 GHz, respectively. Power-controlled amplifiers set the ONU output power P_{Tx}^{US} . Drop and

feeder fibers of length $L_{D-ONU1}=11.4$ km, $L_{D-ONU2}=8.4$ km and $L_F=25$ km are modelled as SSMF with attenuation $\alpha=0.195$ dB/km, dispersion coefficient $D=16.73$ ps/(nm.km) and RBS coefficient -80 dB for a 1 ns pulse width. The US sub-bands are combined after the PS (loss $L_s=23$ dB) and connected to OLT after the feeder fiber. The OLT simultaneously receives the two ONU subcarriers with dual-polarization CoRx, (57 GHz eight-order Butterworth response), trans-impedance amplifiers (TIA) with constant gain and noise, and an 8-bit analog-to-digital converter (ADC) sampling at $f_s=240$ GSa/s. Additional VOAs in the US and DS add loss to match an OPL of 35 dB in the longest link (ONU-1). Together with the receiver noise, the VOAs are parametrized to match the experimental US performance (OLT Rx-DSP, Fig. 1 c)). The OLT Rx-DSP performs subcarrier separation by frequency shift and Gaussian-shaped digital filtering, resampling to 2 samples per symbol (SpS), blind CD compensation, DA-FDE [16], and blind carrier phase estimation (CPE), based on blind phase search algorithm (BPS). In the DS, the OLT broadcasts to the ONUs. The OLT Tx-DSP uses the same initial structure and frame as the ONU Tx-DSP, to generate two DSCM 30 GBd sub-bands with $f_{sep} = 2$ GHz. The OLT Tx-DSP drives an 8-bit DAC, $f_s=120$ GSa/s and DP-IQMs, with 35 GHz fourth-order Butterworth response. A power-controlled amplifier sets the total DS output power $2P_{Tx}^{DS}$. The OLT is connected to the feeder fiber, PS, and the two drop fibers. The ONUs receive their corresponding sub-band with a 57 GHz eight-order Butterworth response CoRx, followed by TIAs with constant gain and

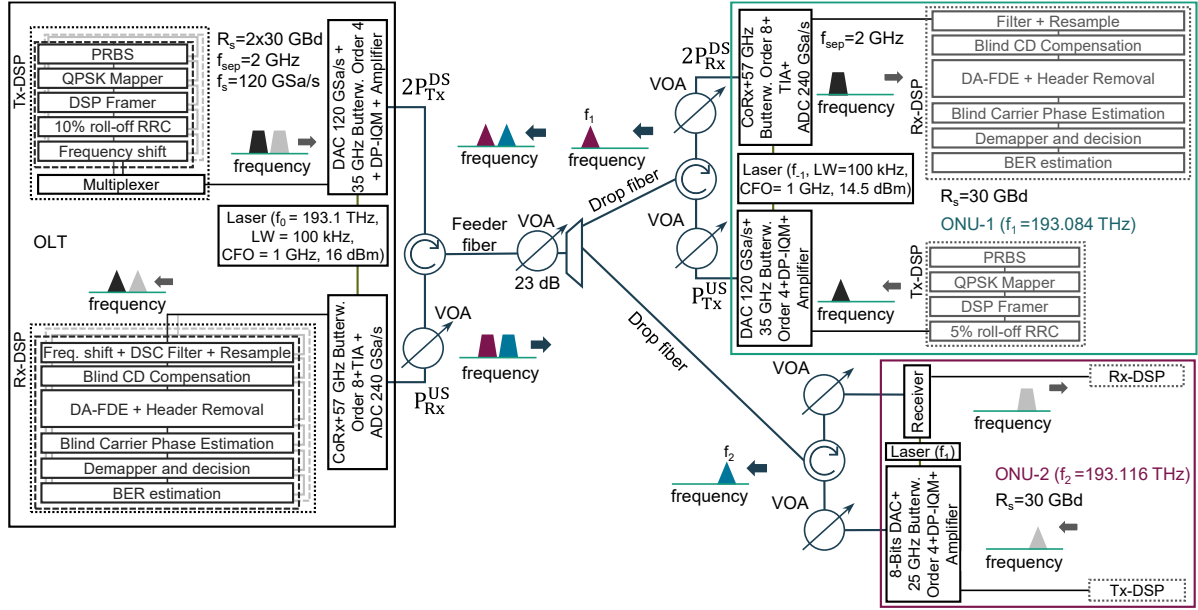


Fig. 2: Simulation setup of bidirectional FDMA coherent PON system, implemented in VPIphotonics™ Design Suite 11.6 using the VPIToolKit™ DSP Library 5.5. $P_{Rx,Tx}^{US,DS}$ Upstream (US) or downstream (DS) received (Rx) or transmitted (Tx) power.

noise, and an 8-bit ADC samples the signal at $f_s=240$ GSa/s. The ONU Rx-DSP performs blind CD compensation, DA-FDE and BPS. Fig. 3 a) and b) show the Q^2 -Factor in the OLT and ONUs receivers, for different combinations of P_{Tx}^{DS} and P_{Tx}^{US} . To analyze the worst-case performance and identify the region where all sub-bands meet the pre-FEC threshold, Fig. 3 reports the minimum Q^2 -factor across all sub bands. In Fig. 3 a), an increase in P_{Tx}^{DS} degrades the US performance at the OLT Rx, whereas increasing P_{Tx}^{US} improves the US performance. In Fig. 3 b), increasing P_{Tx}^{DS} improves the worst-case DS performance at the ONU receivers, but increasing P_{Tx}^{US} degrades the performance due to RBS from the US. In Fig. 3 a) and b) the black line shows the pre-FEC limit of 6.25 dB (BER of $2e-2$ [17]). Fig. 3 c) shows in green the operational region, where the Q^2 -Factor in all the OLT and ONU receivers is above pre-FEC threshold, in red, the regions where pre-FEC threshold cannot be fulfilled. We observe that it is possible to define a combination of P_{Tx}^{DS} and P_{Tx}^{US} that enables bidirectional transmission within the

same wavelength band. The simulation results for the OLT Rx (Fig. 3 a)) reproduce the trends observed in the experimental results (Fig. 2 c)), but with a higher tolerance to P_{Tx}^{DS} , because the model includes only reflections from RBS and neglects connector and circulator reflections. Despite these simplifications, the simulation results isolate the impact of RBS and provides an upper bound on the achievable tolerance.

Conclusions

In this work, we studied the impact of Rayleigh backscattering on a single wavelength DP-QPSK 200 GB/s FDMA coherent PON transmission between two ONUs and an OLT. We experimentally emulate a bidirectional transmission and show the impact of DS reflections in the US link. Based on the experimental investigation, we show by simulations a trade-off between DS (P_{Tx}^{DS}) and US (P_{Tx}^{US}) transmit powers per sub-band. A suitable choice of P_{Tx}^{DS} and P_{Tx}^{US} , enables simultaneous bi-directional transmission in an ODN with 23 dB of PS insertion loss ($Nu=128$) and 35 dB of max. optical path loss (OPL).

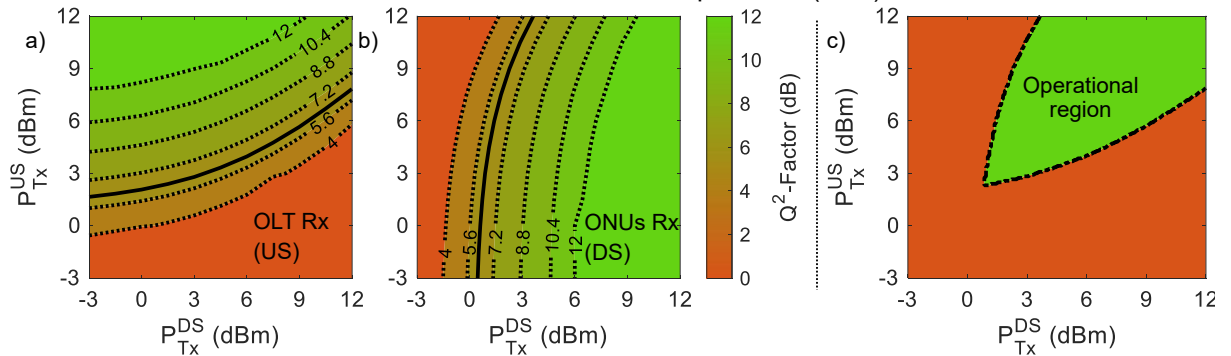


Fig. 3: Simulation results. a), b) worst Q^2 -Factor in the ONU(b) and OLT(c) receivers for different P_{Tx}^{DS} and P_{Tx}^{US} . pre-Black solid line is pre-FEC threshold of 6.25 dB (BER= $2e-2$). c) In green the operational region, where the worst Q^2 -Factor of the OLT and ONUs receivers is higher than the pre-FEC threshold (6.25 dB). In red, the regions where Q^2 -Factor is lower than 6.25 dB.

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